Supplementary Material for Diff-REU: Diffusion Reconstruction Error-Driven Undersampling for Imbalanced Tabular Data Classification

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I. First-Order Convergence Analysis

Let θ denote the trainable parameters of Rec-DDPM. The overall training objective is defined as

$$\mathcal{L}_R(\theta) = \mathcal{L}_{\text{dif}}(\theta) + \lambda_{\text{rec}}\mathcal{L}_{\text{rec}}(\theta), \tag{1}$$

where $\mathcal{L}_{\text{diff}}(\theta)$ denotes the diffusion denoising loss, $\mathcal{L}_{\text{rec}}(\theta)$ denotes the reconstruction loss, and λ_{rec} is the trade-off parameter. In the k-th iteration, the stochastic gradient g_k computed on mini-batch $\mathcal{B}_k$ is

$$g_k = \nabla\mathcal{L}_R(\theta^k; \mathcal{B}_k), \tag{2}$$

and the model parameters are updated by

$$\theta^{k+1} = \theta^k - \eta g_k, \tag{3}$$

where η is the learning rate.

Assumption 1. The objective function $\mathcal{L}_R(\theta)$ is lower bounded, i.e.,

$$\mathcal{L}_R(\theta) \geq \mathcal{L}_R^*, \tag{4}$$

where $\mathcal{L}_R^*$ denotes a finite lower bound of the overall Rec-DDPM objective.

Assumption 2. $\mathcal{L}_{\text{dif}}(\theta)$ and $\mathcal{L}_{\text{rec}}(\theta)$ are L_{dif}-smooth and L_{rec}-smooth, respectively. Thus, the overall objective $\mathcal{L}_R(\theta)$ is L-smooth, where

$$L_R = L_{\text{diff}} + \lambda_{\text{rec}}L_{\text{rec}}. \tag{5}$$

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Assumption 3. The stochastic gradient g_k is unbiased and has bounded variance:

$$\mathbb{E}[g_k|\theta^k] = \nabla\mathcal{L}_R(\theta^k), \tag{6}$$

$$\mathbb{E}\Big[\left\|g_k - \nabla\mathcal{L}_R(\theta^k)\right\|^2 \mid \theta^k\Big] \leq \sigma_R^2. \tag{7}$$

Proposition 1. Under Assumptions 1–3, if the learning rate satisfies $0 < \eta < 1/L$, then after K iterations, Rec-DDPM satisfies

$$\begin{aligned}&\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\Big[\left\|\nabla\mathcal{L}_R(\theta^k)\right\|^2\Big]\\ &\leq \frac{2\left(\mathcal{L}_R(\theta^0) - \mathcal{L}_R^*\right)}{\eta K} + L_R\eta\sigma_R^2.\end{aligned} \tag{8}$$

Furthermore, by choosing $\eta = c/\sqrt{K}$ with a constant $c > 0$ and $\eta < 1/L_R$, we have

$$\begin{aligned}&\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\Big[\left\|\nabla\mathcal{L}_R(\theta^k)\right\|^2\Big]\\ &= O\left(\frac{1}{\sqrt{K}}\right).\end{aligned} \tag{9}$$

Proof. According to the L_R-smoothness of $\mathcal{L}_R(\theta)$, we have

$$\begin{aligned}\mathcal{L}_R(\theta^{k+1}) \leq \mathcal{L}_R(\theta^k) &+ \left\langle\nabla\mathcal{L}_R(\theta^k), \theta^{k+1} - \theta^k\right\rangle\\ &+ \frac{L_R}{2}\left\|\theta^{k+1} - \theta^k\right\|^2.\end{aligned} \tag{10}$$

Substituting $\theta^{k+1} = \theta^k - \eta g_k$ into the above inequality yields

$$\begin{aligned}\mathcal{L}_R(\theta^{k+1}) \leq \mathcal{L}_R(\theta^k) - \eta \left\langle \nabla \mathcal{L}_R(\theta^k), g_k \right\rangle \\ + \frac{L_R \eta^2}{2} \left\| g_k \right\|^2 .\end{aligned} \tag{11}$$

Taking expectation on both sides and using the unbiasedness of g_k, we obtain

$$\begin{aligned}&\mathbb{E}[\mathcal{L}_R(\theta^{k+1})] \\ &\leq \mathbb{E}[\mathcal{L}_R(\theta^k)] - \eta \mathbb{E}\left[\left\| \nabla \mathcal{L}_R(\theta^k) \right\|^2 \right] \\ &\quad + \frac{L_R \eta^2}{2} \mathbb{E}\left[\left\| g_k \right\|^2 \right].\end{aligned} \tag{12}$$

Since g_k can be decomposed into the full gradient and a gradient noise term, we have

$$g_k = \nabla \mathcal{L}_R(\theta^k) + \left(g_k - \nabla \mathcal{L}_R(\theta^k) \right). \tag{13}$$

Then, by the unbiasedness of the stochastic gradient and the variance decomposition, we have

$$\begin{aligned}\mathbb{E}\left[\left\| g_k \right\|^2 \right] =& \mathbb{E}\left[\left\| \nabla \mathcal{L}_R(\theta^k) \right\|^2 \right] \\ &+ \mathbb{E}\left[\left\| g_k - \nabla \mathcal{L}_R(\theta^k) \right\|^2 \right].\end{aligned} \tag{14}$$

According to the bounded-variance assumption, we further obtain

$$\mathbb{E}\left[\left\| g_k \right\|^2 \right] \leq \mathbb{E}\left[\left\| \nabla \mathcal{L}_R(\theta^k) \right\|^2 \right] + \sigma_R^2. \tag{15}$$

Thus, we have

$$\begin{aligned}&\mathbb{E}[\mathcal{L}_R(\theta^{k+1})] \\ &\leq \mathbb{E}[\mathcal{L}_R(\theta^k)] - \left(\eta - \frac{L_R \eta^2}{2} \right) \mathbb{E}\left[\left\| \nabla \mathcal{L}_R(\theta^k) \right\|^2 \right] \\ &\quad + \frac{L_R \eta^2 \sigma_R^2}{2}.\end{aligned} \tag{16}$$

Rearranging the above inequality gives

$$\begin{aligned}&\left(\eta - \frac{L_R \eta^2}{2} \right) \mathbb{E}\left[\left\| \nabla \mathcal{L}_R(\theta^k) \right\|^2 \right] \\ &\leq \mathbb{E}[\mathcal{L}_R(\theta^k)] - \mathbb{E}[\mathcal{L}_R(\theta^{k+1})] + \frac{L_R \eta^2 \sigma_R^2}{2}.\end{aligned} \tag{17}$$

Summing the above inequality from $k = 0$ to $K-1$, we obtain

$$\begin{aligned}&\left(\eta - \frac{L_R \eta^2}{2} \right) \sum_{k=0}^{K-1} \mathbb{E}\left[\left\| \nabla \mathcal{L}_R(\theta^k) \right\|^2 \right] \\ &\leq \mathcal{L}_R(\theta^0) - \mathcal{L}_R^* + \frac{K L_R \eta^2 \sigma_R^2}{2}.\end{aligned} \tag{18}$$

Dividing both sides by $K\left(\eta - \frac{L_R \eta^2}{2} \right)$, we have

$$\begin{aligned}&\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}\left[\left\| \nabla \mathcal{L}_R(\theta^k) \right\|^2 \right] \\ &\leq \frac{\mathcal{L}_R(\theta^0) - \mathcal{L}_R^*}{K\left(\eta - \frac{L_R \eta^2}{2} \right)} + \frac{L_R \eta^2 \sigma_R^2}{2\left(\eta - \frac{L_R \eta^2}{2} \right)}.\end{aligned} \tag{19}$$

When $0 < \eta < 1/L_R$, we have

$$\eta - \frac{L_R \eta^2}{2} \geq \frac{\eta}{2}. \tag{20}$$

Therefore,

$$\begin{aligned}&\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}\left[\left\| \nabla \mathcal{L}_R(\theta^k) \right\|^2 \right] \\ &\leq \frac{2\left(\mathcal{L}_R(\theta^0) - \mathcal{L}_R^* \right)}{\eta K} + L_R \eta \sigma_R^2.\end{aligned} \tag{21}$$

By choosing $\eta = c/\sqrt{K}$ with a constant $c > 0$ and $\eta < 1/L_R$, the right-hand side becomes $O(1/\sqrt{K})$. This completes the proof. ■

II. Sensitivity Analysis

To examine parameter sensitivity across different degrees of class imbalance, we conduct experiments on six datasets: flare_f (IR = 66.80), abalone_20_vs_8_9_10 (IR = 72.69), cleveland_0_vs_4 (IR = 12.62), ecoli_0_1_4_7_vs_2_3_5_6 (IR = 10.59), car_good (IR = 24.04), and page_blocks0 (IR = 8.79). where IR denotes the ratio of majority class to minority class samples. The selected datasets span IR values from 8.79 to 72.69, covering different levels of class imbalance.

A. Parameter Sensitivity

a) Score weights.: This study evaluates five settings for (α, β, γ): $(0.4, 0.3, 0.3)$, $(0.5, 0.25, 0.25)$, $(0.6, 0.2, 0.2)$, $(0.7, 0.15, 0.15)$, and $(0.8, 0.1, 0.1)$. Throughout this experiment, $C_r = 2.0$, $K_r = 1.0$, and $T = 1$ are fixed. As shown in Table I, $(0.4, 0.3, 0.3)$ achieves the highest average AUC of 0.9187 and the best performance on four datasets, the largest count among the evaluated settings. Based on these results, the default weights are set to $\alpha = 0.4$, $\beta = 0.3$, and $\gamma = 0.3$.

b) Candidate pool ratio.: The candidate pool ratio is selected from $C_r \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$. As shown in Table II, $C_r = 2.5$ achieves the highest average AUC of 0.9174. Meanwhile, $C_r = 2.0$ achieves an average AUC of 0.9150 and ties for the largest number of best results across datasets. Since $C_r = 2.0$ yields an average AUC only 0.0024 below the highest value while requiring fewer candidate samples to be evaluated, it is selected as the default.

c) Majority class retention ratio.: The retention ratio is selected from $K_r \in \{0.8, 0.9, 0.95, 1.0, 1.1, 1.2\}$. As shown in Table III, $K_r = 0.8$ achieves the highest average AUC of 0.9185 and the best performance on three datasets, tying with $K_r = 1.1$. Increasing K_r beyond 1.0 retains more majority class samples without improving average AUC. Therefore, the default retention ratio is set to $K_r = 0.8$.

d) Number of ensemble classifiers.: The number of ensemble classifiers is varied over $T \in \{1, 2, \ldots, 15\}$. As reported in Table IV, the optimal value of T differs across datasets. Among the evaluated settings, $T = 7$ is optimal most frequently, achieving the best AUC on two datasets with an average AUC of 0.8679 across those datasets. The settings $T = 1$, 2, 6, and 12 each achieve the highest AUC on one dataset. Based on its most frequent occurrence as the optimal setting, $T = 7$ is selected as the default.

TABLE I
SENSITIVITY ANALYSIS OF WEIGHTING PARAMETERS (α, β, γ).

(α, β, γ)	Avg. AUC ↑	Best Count ↑
(0.4,0.3,0.3)	**0.9187**	**4**
(0.5,0.25,0.25)	0.9108	1
(0.6,0.2,0.2)	0.9125	1
(0.7,0.15,0.15)	0.9081	0
(0.8,0.1,0.1)	0.9121	1

TABLE II
SENSITIVITY ANALYSIS OF CANDIDATE RATIO C_r.

C_r	Avg. AUC ↑	Best Count ↑
1.0	0.9138	**2**
1.5	0.9043	1
2.0	0.9150	**2**
2.5	**0.9174**	1
3.0	0.9150	**2**

TABLE III
SENSITIVITY ANALYSIS OF KEEP RATIO K_r.

K_r	Avg. AUC ↑	Best Count ↑
0.8	**0.9185**	**3**
0.9	0.9057	1
0.95	0.9144	1
1.0	0.9093	1
1.1	0.9149	**3**
1.2	0.9168	1

TABLE IV
AVERAGE BEST AUC ACROSS DATASETS WITH THE SAME OPTIMAL ENSEMBLE SIZE T. VALUES OF T THAT WERE NOT OPTIMAL FOR ANY DATASET ARE OMITTED.

Number T	Average AUC	No. of Datasets
1	**1.0000**	1
2	0.9662	1
6	0.9305	1
7	0.8679	**2**
12	0.9375	1

B. Selection of the Base Classifier

To select the base classifier for Diff-REU, this study evaluates six representative classifiers on the six selected datasets using AUC: Random Forest (RF), Support Vector Machine (SVM), Gradient Boosting Decision Tree (GBDT), Naive Bayes (NB), k-Nearest Neighbors (KNN), and Decision Tree (DT).

As shown in Figure 1, RF achieves the lowest average Friedman aligned rank of 18.25 across the six datasets, followed by GBDT and SVM with ranks of 24.23 and 24.78, respectively. Based on these results, this study selects RF as the default base classifier for Diff-REU.

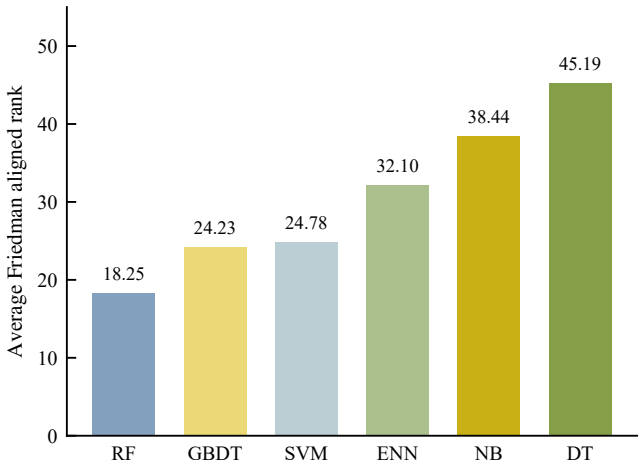


Fig. 1. Average Friedman aligned ranks of six base classifiers across 6 datasets. Lower ranks indicate better overall performance.

III. EXPERIMENTS AND ANALYSIS

A. Performance Comparison of Gini, and TPRGap

Table V reports the Gini and TPRGap results on 14 datasets. Higher Gini indicates stronger discrimination, whereas lower TPRGap indicates a smaller difference between minority class and majority class recall.

Diff-REU achieves the highest Gini on all 14 datasets and the lowest TPRGap on Datasets D, I, L, and N. On Dataset L, Diff-REU reduces TPRGap from 100.00%, the best result among the comparison methods, to 31.70%. On Dataset N, the corresponding values are 100.00% and 20.74%. These results support the discrimination performance of Diff-REU and show that it can also reduce recall disparity on highly imbalanced datasets.

B. Comparison of Alternative Models in Terms of Gini and TPRGap

Table VI reports additional Gini and TPRGap results for five datasets. Rec-DDPM achieves the highest Gini on all five datasets, consistent with its AUC results. On Dataset B, its Gini reaches 75.80%, compared with 74.28% for IF, the strongest alternative on this metric. On Dataset N, the corresponding values are 81.25% and 80.77%.

Different alternative models across datasets achieve the lowest TPRGap. Rec-DDPM remains close to the best result on Dataset E, with a TPRGap of 8.74% compared with 8.62% for NAE. These results support the discrimination advantage of Rec-DDPM, while showing that stronger performance in AUC, G-mean, F1-score, and Gini does not necessarily imply a smaller recall gap between classes.

IV. STATISTICAL TESTS

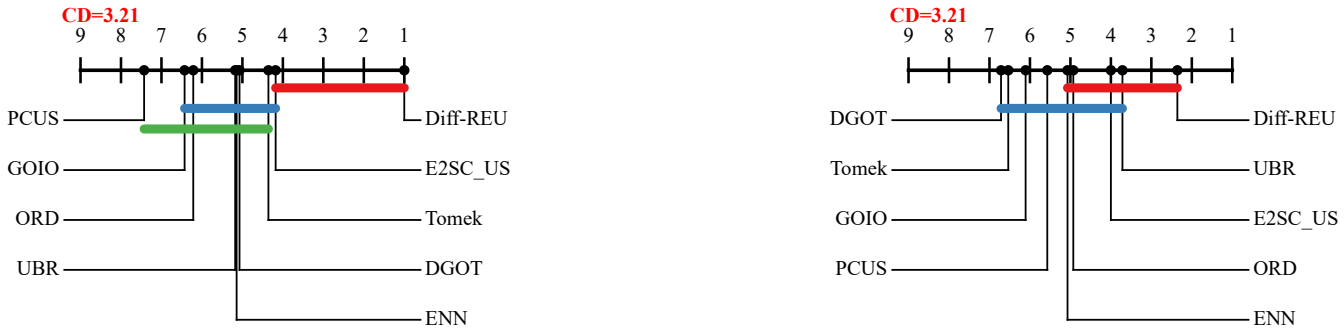
Figure 2 shows the CD diagrams for Gini and TPRGap. Diff-REU ranks first on both metrics and significantly outperforms seven comparison methods on Gini and four on TPRGap.

TABLE V
GINI AND TPRGAP (%) OF DIFF-REU AND EIGHT RESAMPLING METHODS ON 14 DATASETS. HIGHER GINI AND LOWER TPRGAP INDICATE BETTER PERFORMANCE.

Method	Tomek Links	ENN	UBR	E2SC_US	ORD	PCUS	DGOT	GOIO	Diff-REU
Dataset A									
Gini	97.61±0.64	97.45±0.67	97.22±0.82	97.55±0.59	96.38±0.81	97.03±0.76	97.49±0.74	97.49±0.67	**97.68±0.65**
TPRGap	3.61±1.28	2.00±1.18	3.27±2.02	**1.55±1.35**	8.76±2.46	2.49±1.52	3.66±1.79	3.68±1.67	1.61±1.04
Dataset B									
Gini	71.46±9.31	71.56±10.10	67.73±8.93	72.50±8.19	72.35±8.88	46.60±13.56	71.26±10.46	69.38±11.08	**73.81±7.65**
TPRGap	27.27±11.19	15.86±8.40	39.67±9.33	**11.19±8.77**	18.88±7.56	28.19±14.72	42.37±11.43	35.92±16.28	18.74±8.68
Dataset C									
Gini	94.38±10.40	93.75±10.21	95.00±8.74	95.23±7.25	95.03±10.23	93.13±8.56	95.27±5.93	87.19±15.20	**95.31±16.72**
TPRGap	55.63±42.94	55.63±42.94	**11.25±16.61**	30.00±41.44	41.25±44.41	25.63±13.65	69.38±41.71	58.13±42.19	26.25±23.90
Dataset D									
Gini	90.62±12.59	90.41±13.76	89.52±14.83	89.56±15.80	85.76±12.23	72.55±19.32	91.00±16.49	89.52±14.12	**91.07±12.38**
TPRGap	33.01±26.39	33.01±26.39	26.61±21.33	29.03±23.55	20.19±8.07	40.30±13.35	36.99±28.75	25.04±24.02	**18.48±7.05**
Dataset E									
Gini	99.54±0.72	98.72±1.47	99.19±0.70	99.48±0.24	98.68±1.12	83.57±10.14	99.26±1.64	93.67±5.08	**99.58±0.46**
TPRGap	25.89±18.55	16.66±16.46	**4.42±4.21**	10.18±11.38	12.60±1.89	57.81±3.48	41.67±21.45	33.30±21.13	7.23±1.80
Dataset F									
Gini	40.86±7.70	43.63±8.57	46.19±8.47	37.66±7.66	44.89±8.68	34.46±14.95	38.60±9.13	35.94±6.56	**46.82±8.65**
TPRGap	51.44±5.02	7.86±6.28	20.48±9.20	18.62±10.92	**7.58±6.80**	43.63±10.49	62.52±2.53	52.48±9.43	17.92±5.97
Dataset G									
Gini	85.31±1.74	85.66±1.39	85.00±1.23	85.53±1.45	82.16±1.42	67.05±2.59	85.18±1.99	80.72±3.12	**86.05±1.33**
TPRGap	35.60±3.12	19.82±4.45	**3.13±2.09**	17.84±3.41	29.62±2.87	42.12±5.03	37.17±2.99	44.16±4.50	4.13±2.94
Dataset H									
Gini	81.79±2.41	80.55±2.33	82.02±2.24	81.78±2.42	62.84±8.23	81.95±2.66	81.73±2.07	81.97±2.33	**82.23±2.48**
TPRGap	6.20±3.18	27.34±5.66	6.25±3.45	4.08±2.83	87.25±10.78	**3.62±2.44**	4.13±3.02	6.58±5.08	6.43±3.34
Dataset I									
Gini	97.29±3.97	96.99±4.46	96.72±4.92	97.27±3.87	97.16±3.72	94.92±6.64	97.37±3.26	97.27±3.83	**97.68±3.48**
TPRGap	5.32±4.98	5.06±5.82	8.04±7.13	4.73±4.45	5.27±5.24	11.57±9.96	4.95±5.07	5.32±4.98	**4.36±3.97**
Dataset J									
Gini	66.95±19.14	69.27±17.94	72.68±22.93	71.86±15.67	69.16±19.42	79.35±14.34	68.01±17.39	60.99±18.88	**80.04±13.69**
TPRGap	89.63±16.05	55.57±27.71	**15.85±11.71**	73.46±24.43	46.57±26.09	20.00±16.09	91.63±12.46	79.57±24.46	16.78±11.95
Dataset K									
Gini	85.38±15.49	81.78±19.08	83.41±16.93	85.02±14.52	61.25±27.99	86.68±17.60	85.73±13.18	86.04±14.03	**87.72±15.36**
TPRGap	20.19±12.23	19.10±14.29	17.49±10.16	19.23±10.77	84.65±11.88	24.25±12.54	14.44±11.87	**9.54±6.38**	16.40±11.80
Dataset L									
Gini	38.93±30.16	45.90±22.95	48.74±22.21	42.38±26.66	43.59±26.10	28.12±29.36	28.92±23.76	41.18±23.77	**52.01±22.40**
TPRGap	100.00±0.00	100.00±0.00	99.62±0.44	100.00±0.00	99.94±0.20	69.87±12.56	99.87±0.27	96.04±11.87	**31.70±17.24**
Dataset M									
Gini	98.14±1.70	98.24±1.53	97.04±1.59	98.15±1.41	97.25±1.67	91.76±2.69	97.79±1.95	97.84±1.53	**98.37±1.02**
TPRGap	9.29±3.66	5.87±3.25	3.93±2.15	6.04±3.28	**2.54±1.34**	24.33±27.87	10.70±4.33	10.10±3.57	4.71±1.55
Dataset N									
Gini	81.88±16.34	77.22±22.51	81.73±18.37	79.69±23.82	78.69±19.47	76.34±13.22	76.77±22.78	75.30±25.57	**83.02±10.25**
TPRGap	100.00±0.00	94.95±15.79	72.43±25.13	88.17±19.32	87.86±19.37	29.21±19.17	93.07±14.06	84.15±24.19	**20.74±8.57**

TABLE VI
COMPARISON OF REC-DDPM WITH ALTERNATIVE MODELS IN TERMS OF GINI AND TPRGAP (%).

Dataset	Metric	TabDDPM	NAE	IF	GMM	Rec-DDPM
B	Gini↑	73.26±5.06	73.40±3.76	74.28±5.07	73.82±3.92	**75.80±4.40**
	TPRGap↓	16.15±8.06	17.09±8.87	**13.92±8.16**	16.28±8.12	16.77±9.36
D	Gini↑	87.80±8.44	88.38±7.56	87.66±8.09	87.20±7.72	**88.72±6.87**
	TPRGap↓	**6.50±3.07**	8.76±8.33	10.10±7.83	8.54±10.24	8.88±9.98
E	Gini↑	99.28±0.45	99.31±0.51	99.33±0.63	99.50±0.34	**99.70±0.96**
	TPRGap↓	9.22±1.13	**8.62±1.33**	9.94±1.60	9.34±1.23	8.74±0.63
J	Gini↑	77.50±8.79	77.72±6.85	77.92±7.62	77.77±8.57	**78.14±8.54**
	TPRGap↓	15.09±4.94	14.99±5.60	14.68±4.54	**13.92±5.97**	14.50±7.03
N	Gini↑	79.70±9.25	79.58±10.22	80.77±11.00	79.39±9.70	**81.25±9.23**
	TPRGap↓	20.60±7.05	**15.86±11.27**	18.80±8.93	17.90±8.99	18.91±8.64



(a) Gini

(b) TPRGap

Fig. 2. Critical difference (CD) diagrams for Gini and TPRGap. The CD value is shown above each diagram. Lower average ranks indicate better performance, and horizontal lines connect methods whose differences are not statistically significant.